Basic Structural Information of 1HBS:

Information	Description
Protein name	Deoxyhemoglobin S (Hemoglobin S)
PDB ID	1HBS
Organism	Homo sapiens
Oncology Relevance	Not primarily an oncology protein. Hemoglobin S is associated with sickle cell disease, a hereditary blood disorder, rather than cancer.
Mutation	No
Experimental Method	X-ray diffraction
Resolution	3.00 Å
Chain	Chain A
Residue count	141
Bound ligand/cofactor	HEM (heme) – protoporphyrin IX containing Fe

Raw Structure Inspection:
The selected structure, PDB 1HBS, contains eight protein-chain instances (A–H) representing the tetrameric deoxyhemoglobin S structure. There are two unique protein sequences: the α-globin chain, represented by chains A, C, E, and G, and the β-globin chain, represented by chains B, D, F, and H. For single-chain analysis and refinement, Chain A (alpha-globin) can be selected, containing 141 residues. The structure has a total of 1,148 modeled residues across all eight chain instances. The deposited structure is deoxyhemoglobin S from *Homo sapiens* and was determined by X-ray diffraction at 3.00 Å resolution. RCSB PDB reports no engineered mutations for the deposited entities, although hemoglobin S is the sickle-cell form of hemoglobin and its β-globin sequence is associated with the sickle-cell variant. The structure contains one unique ligand, HEM (heme/protoporphyrin IX containing iron), associated with the hemoglobin chains. The biological assembly is a heterotetramer ($\alpha_2\beta_2$)**.**